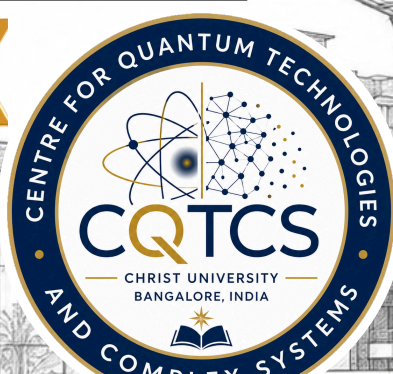
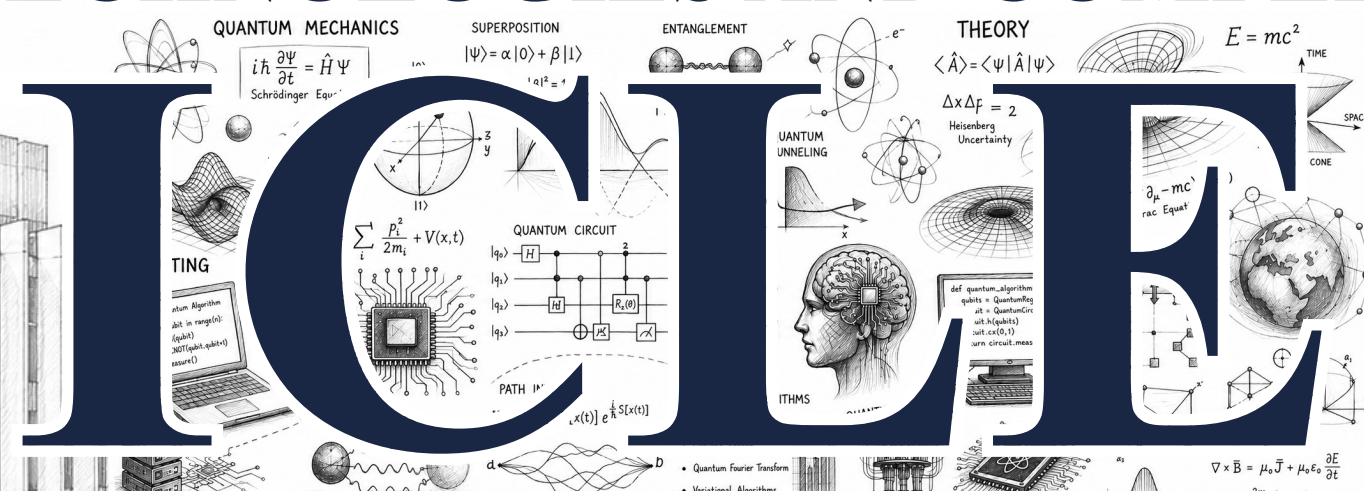




QUANICLE

THE NEWSLETTER



CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE | DELHI NCR | PUNE

COORDINATOR'S NOTE

It is with great pleasure that we present the inaugural issue of QUANICLE, the newsletter of the Centre for Quantum Technologies and Complex Systems (CQTCS).

This platform will showcase our research excellence, celebrate achievements, and highlight opportunities for collaboration in the exciting world of quantum science and complex systems.

Together, let us continue to push the boundaries of knowledge and create a meaningful impact on society.

Dr. Mithun Thudiyangal
Coordinator, CQTCS



REPRESENTING THE CENTRE IN AUSTRIA



12–15 July 2026



Innsbruck, Austria



Ms. Dipasha Chokeda

Mr. Sahil Satapathy

Have been selected to participate in the Introductory Course on Ultracold Quantum Gases, jointly organized by Universität Innsbruck and the Institute for Quantum Optics and Quantum Information (IQOQI), Austria.

The program brings together young researchers from around the world to explore the fascinating physics of ultracold quantum gases.

New Insights into Quantum Superfluid Dynamics

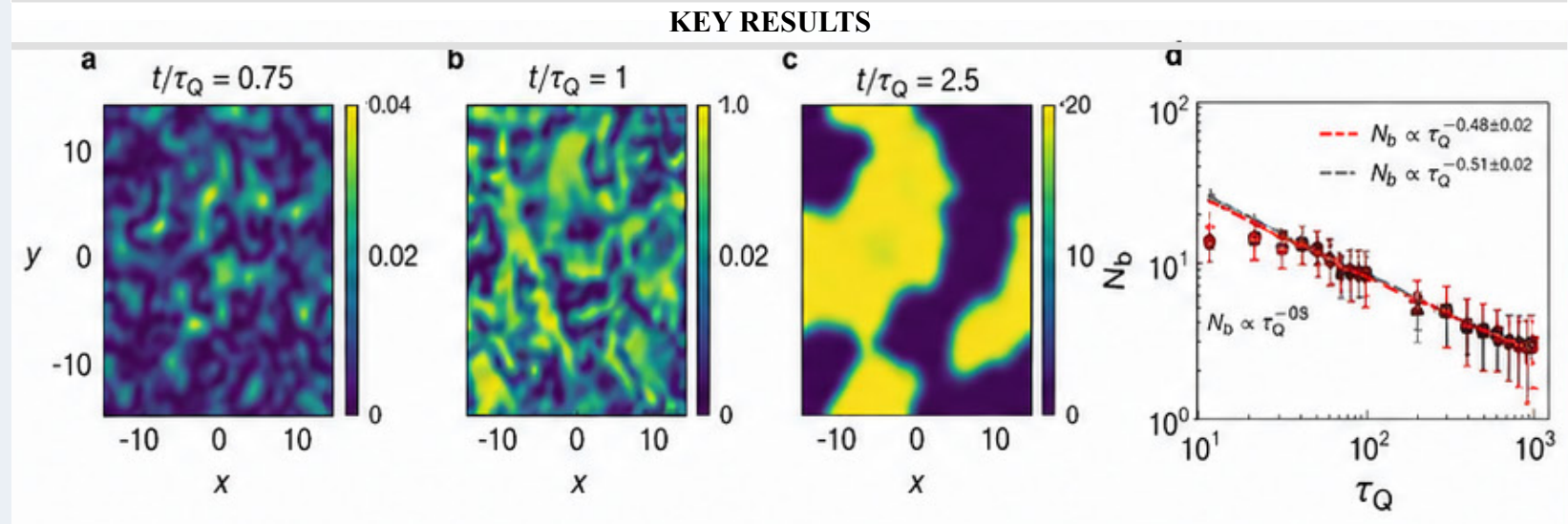
Dr. Mithun Thudiyangal and collaborators from the University of Luxembourg have published their research article, "**Kibble-Zurek Scaling and Spatial Statistics in Quenched Binary Bose Superfluids**" in **Communications Physics (Nature)**, highlighting significant advances in the study of non-equilibrium dynamics in quantum many-body systems. This study explores universal features of far-from-equilibrium dynamics and introduces a framework for characterizing stochastic geometry in multicomponent quantum systems.

communications physics

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Kibble-Zurek scaling and spatial statistics in quenched binary Bose superfluids

Subhadeep Patra¹, Arko Roy¹, Seong-Ho Shinn², Adolfo del Campo^{2,3} & Mithun Thudiyangal⁴



advances in the study of non-equilibrium dynamics in quantum many-body systems. This study explores universal features of far-from-equilibrium dynamics and introduces a framework for characterizing stochastic geometry in multicomponent quantum systems.

PUBLICATIONS

Dr. Sreeja C. S. published a research article in the Q1 journal **IEEE Access**, co-authored with Ph.D. scholar Mr. Vivin Krishnan, titled "**Provably Adaptive Trust Dynamics in Context-Aware Zero-Trust Systems: A Formal Framework for Continuous Verification**"

The paper introduces Zero-Trust Hybrid Adaptive Authentication (ZeTHAA), a continuous authentication and authorization framework integrating contextual attributes, authentication strength, behavioral evidence, and retry dynamics. ZeTHAA utilizes a probabilistic risk model and dual-policy thresholds to partition outcomes into allow, step-up, and block regions, enabling precise control over security-usability trade-offs.

Journals & Magazines > IEEE Access > Volume: 14

Provably Adaptive Trust Dynamics in Context-Aware Zero-Trust Systems: A Formal Framework for Continuous Verification

Publisher: IEEE

Cite This

PDF

Vivin Krishnan & C. S. Sreeja

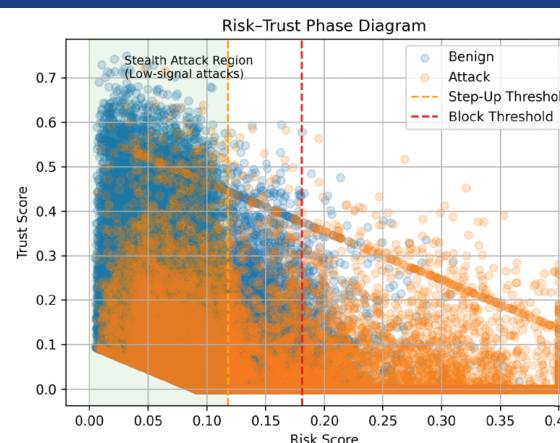
Volume: 14

Quartile: Q1

Impact Factor: 3.9

READ THE PAPER

RISK-TRUST PHASE DIAGRAM



ACADEMIC OPPORTUNITIES

► **M.Sc. Quantum Technologies.** Starting date | 13th July 2026

Advanced postgraduate training in Quantum Computing, Quantum Communication, Quantum Materials, and Quantum Information Science.

► **Certificate Course on Emerging Quantum Technologies.** Starting date | 6th June 2026

Foundational and applied learning in emerging quantum technologies for students, researchers, and professionals.

MEET OUR GROUP LEADERS



Dr. Mithun Thudiyangal

Group Leader



Research Focus

Ultracold Quantum Matter and Quantum Algorithms



Dr. Basudeb Sain

Group Leader



Research Focus

Photonics / Quantum Photonics & Optical Metasurfaces



Dr. Arindam Mallick

Group Leader



Research Focus

Quantum Information, Quantum Simulation & Mathematical Physics



Dr. Sreeja C S

Group Leader



Research Focus

Quantum-Safe Cryptography & Cyber Security



Dr. Ranjith Kumar K M

Group Leader



Research Focus

Quantum Materials & Quantum Magnetism Low Temperature Physics

EVENT RECAP

Dr. Sreeja C S attended the Quantum Pitch Fest, held on 22–23 May 2026 at the Indian Institute of Science (IISc), Bengaluru, organized by WIN-CoE, IISc Bengaluru, IQTI, QuRP, and the Foundation for QC Innovation.



The Centre for Quantum Technologies and Complex Systems (CQTCS), CHRIST (Deemed to be University), was formally inaugurated by **Prof. Subir Sachdev**, Herchel Smith Professor of Physics with the vision of fostering interdisciplinary research and innovation in quantum science, quantum technologies, complex systems, cybersecurity, photonics, quantum materials, and related emerging domains.

